

Automotive Infotainment: Test Challenges and Solutions

Electronics is increasingly being used for in-vehicle infotainment. This opens up a range of challenges at the test implementation stage. Take a look at how these are being overcome by the test industry

SATISH MOHANRAM

The automotive world is moving towards a new level of personalised in-vehicle infotainment. The functionalities involved

at the enterprise level and the test implementation level. Due to the evolution of these standards, there needs to be an integrated framework that maps the product features to the test management

level. At the test engineering level, the biggest challenge faced in the verification and validation phase is the test coverage required to ensure that the systems perform under varied conditions and in various combinations as per the specifications. At the manufacturing phase, the test

and control, and is increasingly being used to test complex electronic systems in automobiles. Software-defined PXI RF test systems provide tremendous benefits for engineers running RF, microwave and wireless test applications. Using key technologies such as field-programmable gate arrays (FPGAs), multi-core CPUs and high-performance RF components, cutting-edge RF test equipment deliver commendable measurement speed, flexibility to create a wide range of applications and cost-effective measurement accuracy.

Fig. 1 shows a modular and software-defined approach to building a test system that is cost-effective, reusable and fast. Test platforms based on such an approach can test several radio standards such as AM, FM, SIRRIUS, DAB, XM, DRM and RDS; navigation standards such as GPS, GLONASS and GALELIO; and mobile communication standards like Wi-Fi 802.11a/b/g/n/ac, Bluetooth, WiMAX, EDGE, CDMA, GPRS and LTE. As these standards evolve, they can still be tested using the same platform. This reduces the system's total cost of ownership (TCO) as the same hardware with changes in software can test multiple standards.

Audio and video test

Digital audio and video are integral parts of an automotive infotainment platform, and their quality greatly influences user satisfaction. Audio, which used to be aired from common speakers, is now becoming personalised for individual passengers, giving

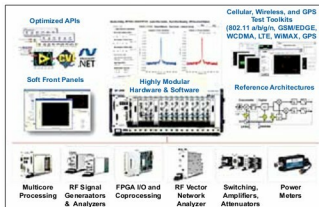


Fig. 1: Multiple radio/navigation standards on one test platform (Courtesy: National Instruments India)

are diverse and can be broadly classified into safety, driver assistance and navigation, personalised interactive multimedia and high-performance data sharing.

These features, in turn, are powered by technological advancements such as sleek 3D-enabled displays, better navigation standards, high-end computational platforms, high-throughput wireless buses, wideband radio communication that enables on-demand high-definition (HD) audio and video on the move, and interactive touch- and gesture-enabled input devices. The best part is that these are evolving and getting better every day.

Such evolutionary car infotainment systems pose a new set of challenges

throughput is a key area of concern. Here is a quick overview of how test engineers are trying to overcome these challenges.

Wireless and radio standards test

There is an increasing demand for providing several radio communication standards on devices, and all within short software development cycles. To cater to this, radio frequency (RF) test engineers adopt well-defined and scalable software architectures. Fast, flexible and accurate RF hardware powered by scalable software can enable effective testing right from design and validation to production.

PXI is an industry-leading, PC-based platform for test, measurement